

LESSON 6-1 • HOMEWORK PACKET • LEVEL 1 • SHEET 1 OF 2**Dividing by a Unit Fraction**

Maryland Standard 6.NS.A.1 · Learning target – I can divide a whole number by a unit fraction and explain what the quotient counts.

REMEMBER

- Dividing by a **unit fraction** asks: how many pieces of that size fit into the number?
- 1 whole holds 4 fourths, so $3 \div \frac{1}{4} = 3 \times 4 = 12$.
- The quotient is **greater** than the number you divide.
Check: $12 \times \frac{1}{4} = 3$.

WORKED EXAMPLE

How many $\frac{1}{3}$ -cup scoops are in 5 cups of rice?

Step 1 Each cup holds 3 thirds.

Step 2 5 cups hold $5 \times 3 = 15$ thirds, so $5 \div \frac{1}{3} = 15$.

Step 3 Check: $15 \times \frac{1}{3} = 5$, so it checks.

Answer 15 scoops

WORD BANK

unit fraction – a fraction with 1 on top, like $\frac{1}{4}$

reciprocal – the flipped number: $\frac{1}{4}$ becomes 4

quotient – the answer when you divide

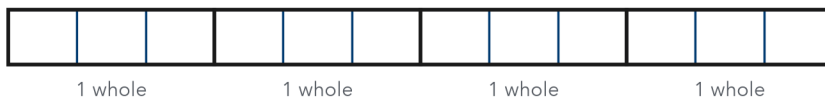
PRACTICE

show your work · label units

1. Each bar is 1 whole. Use the fraction bars to find $4 \div \frac{1}{3}$. How many thirds are in 4 wholes?

Step 1 Count the thirds in 1 bar.

Step 2 Count the thirds in all 4 bars.



$4 \div \frac{1}{3} =$ _____

2. Divide.

a. $6 \div \frac{1}{2} =$ _____

b. $3 \div \frac{1}{5} =$ _____

c. $8 \div \frac{1}{4} =$ _____

d. $2 \div \frac{1}{9} =$ _____

e. $7 \div \frac{1}{3} =$ _____

f. $10 \div \frac{1}{6} =$ _____

g. $5 \div \frac{1}{7} =$ _____

h. $12 \div \frac{1}{3} =$ _____

3. **Trail Mix.** Jamal has **6 cups** of trail mix. Each snack bag holds **$\frac{1}{4}$ cup**. How many bags can he fill?

Step 1 Write the division: $6 \div \frac{1}{4}$.

Step 2 Change it to multiplication: 6×4 .

Answer _____ bags

- 4. Ribbon Bows.** Ms. Carter has **5 yards** of ribbon. She cuts it into pieces that are **$\frac{1}{6}$ yard** long. She **uses 18** pieces. How many pieces are **left**?

Step 1 First find $5 \div \frac{1}{6}$.

Step 2 Then subtract the 18 pieces she uses.

Show your work.

Answer _____ pieces

- 5. Find the Error.** Tanya says $8 \div \frac{1}{2} = 4$. Explain her mistake. Then find the correct quotient.

Frame Tanya found ____ of 8. The correct answer is ____ because ____.

CHECK YOURSELF

☐ I can do this on my own

☐ I need a little help

☐ I have a question about #____

AT HOME Count how many $\frac{1}{4}$ -cup scoops fill 2 cups. Does it match $2 \div \frac{1}{4}$?

LESSON 6-1 • HOMEWORK PACKET • LEVEL 1 • SHEET 2 OF 2**Dividing a Fraction by a Whole Number**

Maryland Standard 6.NS.A.1 · Learning target – I can divide a fraction by a whole number and explain the result.

REMEMBER

- Dividing by a whole number **shares** a fraction into equal parts.
- Multiply by the reciprocal: $\frac{2}{3} \div 4 = \frac{2}{3} \times \frac{1}{4} = \frac{2}{12} = \frac{1}{6}$.
- Dividing by a whole number greater than 1 gives a quotient **less** than the fraction. Check: $\frac{1}{6} \times 4 = \frac{2}{3}$.

WORKED EXAMPLE3 friends share $\frac{3}{4}$ of a sandwich equally. What part of the whole does each get?**Step 1** $\frac{3}{4} \div 3 = \frac{3}{4} \times \frac{1}{3} = \frac{3}{12}$ **Step 2** Simplify: $\frac{3}{12} = \frac{1}{4}$ **Answer** $\frac{1}{4}$ of the sandwich**WORD BANK****share** – split into equal parts**reciprocal** – the flipped number: 6 becomes $\frac{1}{6}$ **simplest form** – divide top and bottom by the same number**PRACTICE***show your work · label units*

- 1.**
- Divide. Write each quotient in simplest form.

a. $\frac{1}{2} \div 3 =$ _____

b. $\frac{4}{5} \div 2 =$ _____

c. $\frac{3}{8} \div 3 =$ _____

d. $\frac{2}{3} \div 6 =$ _____

e. $\frac{6}{7} \div 2 =$ _____

f. $\frac{5}{6} \div 10 =$ _____

- 2.**
- Find each share. Write it in simplest form.

a. 4 people share $\frac{1}{2}$ lb of cheese: _____ lb

b. 5 cups share $\frac{5}{6}$ L of juice: _____ L

- 3. Paint Cans.**
- Kiara has
- $\frac{3}{5}$
- gallon of paint. She pours it
- equally**
- into
- 6 cans**
- . How much paint is in each can?

Step 1 Write the division: $\frac{3}{5} \div 6$.**Step 2** Multiply $\frac{3}{5}$ by $\frac{1}{6}$.**Step 3** Simplify your answer.*Show your work.***Answer** _____ gallon

4. **Clay.** Mr. Okafor has $\frac{2}{3}$ pound of clay. He splits it equally among 4 students. How much clay does each student get?

Step 1 Write the division: $\frac{2}{3} \div 4$.

Step 2 Multiply $\frac{2}{3}$ by $\frac{1}{4}$, then simplify.

Answer _____ pound

5. **Reason.** Without dividing, decide: is $\frac{5}{8} \div 4$ greater than or less than $\frac{5}{8}$?

Frame $\frac{5}{8} \div 4$ is ____ than $\frac{5}{8}$ because ____.

☐ greater than $\frac{5}{8}$

☐ less than $\frac{5}{8}$

TEST PRACTICE

choose the best answer

6. Which expression has the same value as $\frac{5}{6} \div 5$?

A $\frac{5}{6} \times 5$

B $\frac{5}{6} \times \frac{1}{5}$

C $\frac{6}{5} \times 5$

D $\frac{5}{6} - 5$

7. **4 people** share **$\frac{4}{9}$** of a pizza **equally**. What fraction of the whole pizza does each person get?

A $\frac{1}{9}$

B $\frac{16}{9}$

C $\frac{1}{36}$

D $\frac{4}{9}$

CHECK YOURSELF

☐ I can do this on my own

☐ I need a little help

☐ I have a question about #____

AT HOME Share $\frac{1}{2}$ of a sandwich equally among 2 people at home. What fraction of the whole sandwich does each get?